Leetcode 1392 Longest Happy Prefix

Tuesday, March 10, 2026 6:21 AM

```python
class Solution:
    def longestPrefix(self, s: str) -> str:
        n= len(s)
        for i in range(n-1, 0 ,-1):
            if s[:i] == s[n-i:]:
                return s[:i]
        return ""
```

Bruteforce

s = "abaabab"

n = 6

Check Prefix of Length i from n-1 → 1

Prefix → s[0:i]

Suffix → s[n-i:n]

s+e) 1

i = 5

Prefix s[0:5] = "abaaba"

Suffix

" $S[6-5:6] = S[1:6]$
= " bab ab "

Compare
" ab aba " $\neq$ " b abab "

Not Equal

Step 2
$i = 4$

Prefix

$S[0:4]$ = "abab"

Suffix
$S[6-4:6] = S[2:6]$ = "abab"

Compare,
" abab " = " abab "

match found

Return
" abab "

Brute Force Visualization

i	Prefix	Suffix	Match
5	ababa	babab	×

4 abab abab ✓

Time Complexity -> $O(n^2)$

Space Complexity $O(1)$

Optimal

```python
class Solution:
    def longestPrefix(self, s: str) -> str:
        n= len(s)
        lps=[0] *n

        length=0
        i=1

        while i < n:
            if s[i] == s[length]:
                length +=1
                lps[i]= length
                i+=1
            else:
                if length !=0:
                    length = lps[length -1]
                else:
                    lps[i]=0
                    i+=1
        return s[:lps[-1]]
```

S = "abaabab"
n = 6

LPS[i] = Longest Prefix == Suffix for Substring [0 .. i]

for
S = "abaabab"
Initialize

lps = [0,0,0,0,0,0]
Length = 0
i = 1

Step by Step

i = 1
Compare
S[1] = 'b'
S[length] = S[0] = 'a'

Not Equal
lps[1] = 0
i = 2

i = 2

LPS
[0,0,0,0,0,0]

i = 2

Compare
S[2] = 'a'
S[length] = S[0] = 'a'

Match
length = 1
lps[2] = 1
i = 3

LPS
[0,0, 1,0,0,0]

i = 3

Compare
S[3] = 'b'
S[length] = S[1] = 'b'

Match
length = 2

lps[3] = 2
i = 4

LPS
[0, 0, 1, 2, 0, 0]

i = 4

Compare
S[4] = 'a'
S[length] = S[2] = 'a'

Match
length = 3
lps[4] = 3
i = 5

LPS
[0, 0, 1, 2, 3, 0]

i = 5

Compare,
S[5] = 'b'
S[Length] = S[3] = 'b'

Match
length = 4
lps[5] = 4

i = 6

LPS

[0, 0, 1, 2, 3, 4]

Final LPS Array

index :	0	1	2	3	4	5
Char :	a	b	a	b	a	b
lps :	0	0	1	2	3	4

Last Value

lps[n-1] = 4

Answer

s[:4] : 'abab'

Time Complexity O(n)

Space Complexity O(n)